

MENOMINEE HARBOR, MI, USA

WMO: 997785

Lat: 45.0960N

Lon: 87.5900W

Elev: 584

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.8	(c) -1.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 28.7	(k) 20.9	(l) 26.3	(m) 21.2	(n) 9.3	(o) 270	(p) 0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.6	(c) 83.5	(d) N/A	(e) 80.4	(f) N/A	(g) 77.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 10.6	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 26.6	(b) 23.5	(c) 20.2	(e) -10.6	(f) 89.7	(g) 6.2	(h) 3.0	(i) -15.1	(j) 91.9	(k) -18.7	(l) 93.6	(m) -22.2	(n) 95.3	(o) -26.7	(p) 97.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	18.1	18.7	28.9	40.6	53.1	63.7	69.9	68.5	60.6	49.5	36.5	25.7
	DBStd	19.96	11.12	10.44	10.50	7.14	7.42	6.30	5.53	4.66	6.98	7.45	8.93	9.26
	HDD50	4133	989	877	654	296	50	0	0	0	4	100	410	753
	HDD65	7836	1454	1297	1118	733	375	98	18	18	169	484	854	1218
	CDD50	2174	0	0	1	13	146	410	618	574	321	85	6	0
	CDD65	402	0	0	0	0	6	58	171	127	36	4	0	0
	CDH74	1625	0	0	0	3	29	231	819	424	108	11	0	0
	CDH80	295	0	0	0	1	3	33	182	62	13	1	0	0
(14) Wind	WSAvg	10.1	10.4	10.7	9.6	11.2	10.7	9.4	8.8	8.5	9.8	10.9	11.0	10.8
(15) Precipitation	PrecAvg	30.0	1.5	1.0	1.7	2.6	3.5	3.9	3.5	3.0	3.2	3.0	2.0	1.6
	PrecMax	37.7	5.4	2.2	4.0	4.7	5.7	8.8	6.5	5.6	6.2	7.1	5.7	3.5
	PrecMin	23.0	0.4	0.2	0.4	0.7	1.4	1.3	1.1	0.5	1.3	0.4	0.2	0.2
	PrecStd	3.6	1.1	0.5	0.9	1.2	1.2	2.2	1.5	1.3	1.4	1.7	1.3	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	43.2	60.0	65.7	78.1	84.6	89.6	85.6	81.5	74.6	59.9	45.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	39.6	38.5	52.3	58.6	72.4	80.1	84.4	81.6	77.0	67.7	54.8	41.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	36.6	35.4	46.5	53.8	68.2	76.7	81.2	78.3	73.5	63.2	51.3	39.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	33.7	33.0	42.2	50.2	64.8	73.2	78.5	76.0	70.9	60.4	48.4	37.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	13.5	13.9	10.9	12.1	11.5	12.6	12.0	12.3	10.3	9.6	10.0
		MCDBR	12.7	13.5	17.4	17.6	18.4	16.8	15.6	14.4	15.3	13.4	11.2	9.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.261	0.290	0.329	0.360	0.392	0.395	0.386	0.388	0.364	0.330	0.300	0.271
	taud		2.463	2.363	2.371	2.357	2.323	2.351	2.393	2.381	2.442	2.527	2.529	2.494
	Ebn at Noon		277	285	287	286	278	277	278	274	272	268	259	261
	Edh at Noon		22	29	33	37	39	38	36	36	31	25	21	19
(44) All-Sky Solar Radiation	RadAvg		450	767	1148	1447	1758	1900	1953	1658	1283	789	469	345
	RadStd		47	83	106	136	141	146	82	110	96	81	45	35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page